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Roll No. : .....

5100131(024)

M. Tech. (First Semester) Examination 2019

(Old Scheme)

Elect. Engineering

Specialization : High Voltage Engineering

Power Electronics for Renewable Sources

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 40

Note : *All questions are compulsory and carry equal marks. Solve any* ***two*** *part from (a), (b) and (c) of each unit.*

Unit - I

1. (a) What is a fuel cell? Mention the different types of fuel cell and explain any two of them in detail with neat diagram.

(b) Explain the construction, working and different characteristics of solar array in detail.

(c) List out the available renewable energy sources. Explain how solar and wind energy sources plays significant role of electric power generation.

Unit - II

2. (a) Discuss in detail about the construction and working of permanent magnet synchronous generator.

(b) Draw the equivalent circuit and show the steady state analysis of Permanent Magnet Synchronous generator (PMSG). https://www.csvtuonline.com

(c) Draw the equivalent circuit and obtain the steady state analysis of Induction Generator.

Unit - III

3. (a) Explain the different modes of operation of PV Fed Buck-Boost converter in detail.

(b) Explain the need of AC-DC-AC converters for wind energy conversion system.

(c) What is a Matrix Converter? Explain it in detail.

Unit - IV

4. (a) Write a detailed note on stand alone operation of photovoltaic system.

(b) Draw the general structure of variable speed wind energy conversion for stand alone system. Explain the function of components used.

(c) Explain the circuit model of grid integrated solar system.

Unit - V

5. (a) Explain any three different configuration of Hybrid renewable energy system.

(b) Explain the operation of autonomous ϕv with an MPPT converter and battery backup with neat sketch.

(c) What is called Maximum Power Point Tracking (MPPT)? Explain the use of MPPT for hybrid wind and photovoltaic energy system.